

Briefing for Semester 2024 September

CSC3034 Computational Intelligence

Dr Tang Tiong Yew

My info

Dr Tang Tiong Yew

Current contact details are available on the CSC3034 course site.

Department of Computing and Information Systems

By the end of the subject, the students would be able to

- ▶ **CLO1** implement appropriate CI-based solution on typical CI problems,
- ▶ **CLO2** implement appropriate hybrid CI system to solve a computational problem, and
- ▶ **CLO3** evaluate quality of CI solutions using appropriate performance metrics.

	Weightage	CLO1	CLO2	CLO3
Group Assignment 1 (Fuzzy systems)	20	✓		
Group Assignment 2 (Hybrid systems)	10		✓	
Quiz (Hybrid systems, Performance metrics)	20		✓	✓
Final Examination	50	✓	✓	✓

Lecture Plan

Week	Lecture (Fri)	Date	Practical (Mon/Thu)	Remark
1 20/09	Introduction	16,19/09	-	16-17/09 Malaysia Day
2 27/09	-	23,26/09	Python refresh	
3 04/10	[FZ] Fuzzy logic + Fuzzy sets	30/9,03/10	-	
4 11/10	[FZ] Fuzzy inference systems	07,10/10	Fuzzy systems	11/10 Asgm 1 briefing
5 18/10	[EC] Genetic algorithms	14,17/10	-	
6 25/10	[EC] Particle swarm optimisation + Ant colony optimisation	21,24/10	Genetic algorithms	
28/10 - 01/11 Mid-sem break				31/10 Deepavali
7 08/11	[ANN] Artificial neural networks	04,07/11	Particle swarm optimisation	08/11 Asgm 1 deadline
8 15/11	[ANN] Unsupervised learning with ANN	11,14/11	Ant colony optimisation	
9 22/11	[ANN] Recurrent neural networks	18,21/11	Supervised learning with ANN	22/11 Asgm 2 briefing
10 29/11	Hybrid systems	25,28/11	Hyperplane in ANN	
11 06/12	Performance metrics	02,05/12	ANN with Keras	06/12 Asgm 2 deadline
12 13/12	Summary + Revision	09,12/12	Performance metrics	11/12 Sultan of Selangor's Birthday
13 20/12	Quiz	16,19/12	-	
14 27/12	-	23,26/12	-	25/12 Christmas Day

FZ Fuzzy systems

EC Evolutionary computations

ANN Artificial neural networks

Lectures

Friday 10:00 – 12:00 • Auditorium 2

Labs

Face-to-face sessions: Dr Tang Tiong Yew

Lab sheets: <https://robotictang.github.io/CSC3034/docs/>

Materials through elearn

Microsoft Teams

Post in respective channel

Lectures

Face-to-face sessions for practicals

Session 1: Thursday · 10:00 – 12:00 (UW-2-8)

Session 2: Monday · 10:00 – 12:00 (SW-CL1-9)

Session 3: Thursday · 08:00 – 10:00 (UW-2-8)

1. Lucci, S, & Kopec, D. (2016). Artificial Intelligence in the 21st Century - 2nd ed. Mercury Learning and Information.
2. Negnevitsky, M. (2005). Artificial Intelligence: A Guide to Intelligent Systems - 2nd ed. Addison Wesley.
3. Engelbrecht, A. P. (2007). Computational Intelligence: An Introduction - 2nd ed. Wiley.
4. Konar, A. (2005). Computational Intelligence: Principles, Techniques and Applications. Springer.
5. Rutkowski, L. (2010). Computational Intelligence: Methods and Techniques. Springer.
6. Eberhart, R. & Shi, Y. (2007). Computational Intelligence: Concepts to Implementations. Morgan Kaufmann, Elsevier.

CSC3034 Current Course Update

00 Briefing

Synchronized from the current CSC3034 course site on 14 August 2026.

- Current delivery: eight practicals scheduled from Week 2 to Week 12.
- Isaac Sim 6.0 uses workstation archives; Omniverse Launcher is retired.
- CPU/Matplotlib fallbacks remain available when an RTX workstation is unavailable.

Current materials author: Dr Tang Tiong Yew

See the synchronized lab notes and runnable examples for full instructions.